MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 1B24A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR, tunable, internal cavity, incident power 30 kw, frequency range 8,490 to 9,600 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Ii	Ebb	Alt
Unit:	kw	μAdc	Vdc	ft
Maximum:	30	200	-1,000	10,000
Minimum:	---	100	-750	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1

TEST CONDITIONS:

Parameter:	Incident power	Ii	tp	prr	Ebb	Ri
Unit:	kw	μAdc	μs	pps	Vdc	Meg
Test condition 1:	10	100	0.5	1,000	---	---
Test condition 2:	30	---	1.0	1,000	-800	2.3

Frequency		
F	MHz	±
F1	8,490	0.1%
F2	8,600	0.1%
F3	9,000	0.1%
F4	9,375	0.1%
F5	9,375	0.2%
F6	9,375	0.5%
F7	9,500	0.1%
F8	9,600	0.1%

GENERAL:

Qualification - Required

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location of guide lug or control grid contact shall be referenced by a notch or arrow on the anode radiator in position

METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification							
4416	Insertion loss (integral cavity)	2	---	F = F2 to F7	Li	---	2.0	dB
4461	Loaded Q	2	---	F = F2 to F7	QL	160	350	---
4466	Temperature coefficient of frequency	3	---	F = F4	F	---	-20	MHz
	Quality conformance inspection, part 1	1	---					
4401	Ignitor ignition time	---	---	Ebb = -800 Vdc; Ri = 2.3 Meg ±1%	t	---	5.0	sec
4406	Ignitor voltage drop	---	---	Ii = 100 μAdc	Eid	325	450	Vdc
4416	Insertion loss (integral cavity)	---	---	F = F1	Li	---	2.0	dB
4421	Ignitor interaction (insertion loss)	---	---	Ii = 100 μAdc	ΔLi	---	0.2	dB
4426	Tuning range	---	---	5 turns	F	8,490	9,600	MHz
4452	Flat leakage power	---	1	F = F6; T = 25° ±5°C	pf	---	30	mw
1027	Temperature cycling (nonoperating)	---	---		---	---	---	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	---	---		---	---	---	---
4021	Degradation due to vibration	---	---		---	---	---	---
4411	Ignitor oscillation	---	---		Ii	---	60	μAdc
4416	Insertion loss (integral cavity)	---	---	F = F3, F4, and F8	Li	---	2.0	dB
4461	Loaded Q	---	---	F = F1 and F8	QL	160	350	---
4471	Recovery time	---	1	F = F4	t	---	4.0	μs
4016	Vibration detuning	4	---	2G; F = F4; t = 12 hours	ΔF	---	±3	MHz
	Quality conformance inspection, part 3							
---	Life test	5	2	Group B; F = F5	t	500	---	hrs

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 - Continued							
---	Life-test end points:	6	---		---	---	---	---
4452	Flat leakage power	---	1	F = F6; T = 25° ±5°C	pf	---	30	mw
4416	Insertion loss (integral cavity)	---	---	F = F4	Li	---	2.0	dB
4421	Ignitor interaction (insertion loss)	---	---	Ii = 100 μAdc	ΔLi	---	0.5	dB
4406	Ignitor voltage drop	---	---	Ii = 100 μAdc	Eid	---	650	Vdc
4021	Degradation due to vibration	---	---		---	---	---	---
4471	Recovery time	---	1	F = F4	t	---	6	μs
1027	Temperature cycling life test	---	---	Group B	Cycles	10	---	---

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.
2. This test shall be performed at intervals of 100 MHz.
3. The frequency drift shall be measured with no adjustment of the tuning mechanism.
4. The tube shall be tuned to the specified frequency and then vibrated in the direction of ignitor axis. At the conclusion of this test, the tube shall satisfy all electrical tests of this specification.
5. The ignitor current shall not be adjusted during life test.
6. Life-test end points shall be measured by using a fixed voltage and resistor.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11, 17
DSA - ES

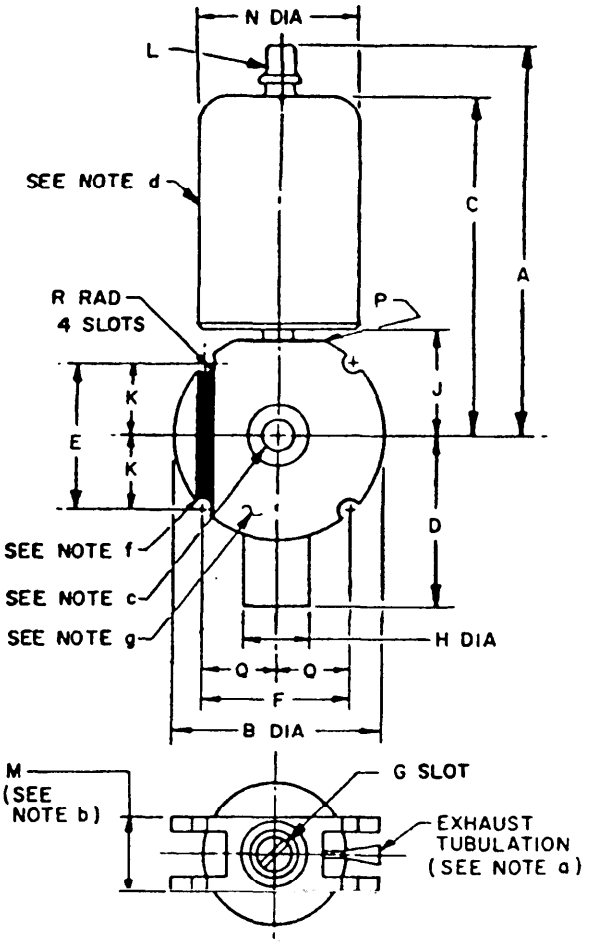
User activities:
Army - MU, WC, AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
C	---	2.875 (73.03)
E	1.275 (32.39)	1.285 (32.64)
F	1.215 (30.86)	1.225 (31.12)
L	Cap: C1-2 (EIA)	
Quality conformance inspection, part 1 (see note 1)		
G	.031 (.79)	.063 (1.60)
K	.637 (16.18)	.643 (16.33)
M	.609 (15.47)	.615 (15.62)
Q	.607 (15.42)	.613 (15.57)
R	.086 (2.18)	.094 (2.39)
Quality conformance inspection, part 2		
A	---	3.250 (82.55)
B	---	1.760 (44.70)
D	---	1.422 (36.12)
H	.527 (13.39)	.533 (13.54)
J	.938 (23.83)	---
N	---	1.188 (30.18)
Reference dimension		
P	.563 (14.30) Flat	

NOTES:
a. Exhaust tubulation shall not extend beyond periphery.
b. Applies to area between periphery of this section of tube and concentric circle of 5/16 radius.
c. No part of iris assembly shall extend beyond body surface.
d. Reservoir shall be glass, or approved equivalent.
e. Maximum projection of reservoir shall lie within a cylinder of 1.250 diameter with axis colinear with tube axis.
f. A force of 200 pounds shall be applied to the face of the tube within the area indicated by shading. Dimension M shall not be permanently changed by more than 0.001. Qualification inspection required.
g. Body faces shall be cadmium plated 0.0003 minimum, or shall be made entirely of monel, or equivalent.
h. Solder fillets permissible on peripheral surface near exhaust tubulation and electrode terminal. Slots shall be free from solder.

FIGURE 1. Outline drawing of electron tube type 1B24A.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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